Operating Systems
Assignment 4 – *Easy*

Instructions:

1. The assignment has to be done solo.
2. You can use Piazza for any queries related to the assignment and avoid asking queries on the last day.
3. There will be a live demonstration and viva to evaluate the fourth assignment.

1 Goals

Implement a Docker container in xv6. The assignment statement is deliberately kept short because it is open-ended. There will be a viva. The Docker-like container should support :

- Job suspension
- Checkpointing
- Job migration
- Virtual file system
- At least one virtual device

Note: It should be possible to run multiple such containers together.

2 Submission Instructions

- We will run MOSS on the submissions. Any cheating will result in a zero in the assignment, a penalty as per the course policy, and possibly much stricter penalties (including a fail grade).
- Create a report file with the name A4_report.pdf that clearly specifies your understanding of the Docker container and explains the design choices during implementation.

How to submit:

1. Copy your report to the xv6 root directory.

2. Then, in the root directory, run the following commands:

```
make clean
tar czvf \
  assignment4_easy_<entryNumber1>.tar.gz *
```

   This will create a tarball with the name, $assignment3\_easy\_ < entryNumber1 > .tar.gz$ in the same directory that contains all the xv6 files and the report PDF document. Entry number format: 2020CSZ2445 (*All English letters will be in capitals in the entry number.*).

3. Please note that if the report is missing in the root directory, no marks will be awarded for the report.